

Full System Database Roadmap - All 8 Tables

1. CUSTOMERS TABLE (Consignors & Consignees)

Connects to LrBilty.

Column Name	Data Type	Constraint	Null Rules & Details
CustomerID (PK)	INT	NOT NULL	Auto-increment primary key.
CustomerType	VARCHAR(20)	NOT NULL	Consignor, Consignee, or Broker.
FullName	VARCHAR(100)	NOT NULL	Registered business/person name.
GSTIN	VARCHAR(15)	NULL ALLOWED	Empty for unregistered, Fill for B2B.
ContactPhone	VARCHAR(15)	NOT NULL	Primary contact number.
Address	TEXT	NOT NULL	Complete physical address.

2. VEHICLES TABLE

Connects to LrBilty, EWayBill, and Tracking.

Column Name	Data Type	Constraint	Null Rules & Details
VehicleID (PK)	INT	NOT NULL	Auto-increment primary key.
RegistrationNo	VARCHAR(20)	NOT NULL	Number plate (e.g., MH12AB1234).
OwnerName	VARCHAR(100)	NOT NULL	Name of the vehicle owner / agency.
VehicleType	VARCHAR(30)	NOT NULL	Open Body, Container, Trailer, etc.
LoadCapacity	DECIMAL(10,2)	NOT NULL	Max weight truck can carry.
RCExpiryDate	DATE	NULL ALLOWED	Empty if not tracked. Fill to prevent dispatch.

3. DRIVERS TABLE

Connects to LrBilty.

Column Name	Data Type	Constraint	Null Rules & Details
DriverID (PK)	INT	NOT NULL	Auto-increment primary key.
DriverName	VARCHAR(100)	NOT NULL	Driver full name.
LicenseNumber	VARCHAR(30)	NOT NULL	Official driving license code.
LicenseExpiry	DATE	NOT NULL	Date when the license expires.
Phone	VARCHAR(15)	NOT NULL	Primary contact number.

4. USERS / STAFF TABLE

Connects to LrBilty (Created by) & Tracking (Updated by).

Column Name	Data Type	Constraint	Null Rules & Details
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UserID (PK)	INT	NOT NULL	Auto-increment primary key.
Username	VARCHAR(50)	NOT NULL	Login name.
PasswordHash	VARCHAR(255)	NOT NULL	Securely hashed password.
Role	VARCHAR(20)	NOT NULL	Admin, Manager, Clerk.
IsActive	BOOLEAN	NOT NULL	Active/Inactive Status.

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5. INVOICES & PAYMENTS TABLE

Independent ledger for Advance/Balance amounts. Connects to LrBilty.

Column Name	Data Type	Constraint	Null Rules & Details
InvoiceID (PK)	INT	NOT NULL	Auto-increment primary key.
LrNumber (FK)	VARCHAR(30)	NOT NULL	The Bilty this payment belongs to.
PaymentType	VARCHAR(20)	NOT NULL	Advance Payment or Balance Payment.
AmountPaid	DECIMAL(10,2)	NOT NULL	Exact money received.
PaymentDate	DATETIME	NOT NULL	Exact time of transaction.
TransactionRef	VARCHAR(100)	NULL ALLOWED	Empty for Cash. Fill the Bank UTR for online.

6. TRACKING TABLE

History timeline logs. Connects to LrBilty.

Column Name	Data Type	Constraint	Null Rules & Details
TrackingID (PK)	INT	NOT NULL	Auto-increment primary key.
LrNumber (FK)	VARCHAR(30)	NOT NULL	Connects back to the Bilty.
CurrentLocation	VARCHAR(200)	NOT NULL	Specific checkpoint / city name.
Status	VARCHAR(50)	NOT NULL	Created, CheckpointReached, Delivered.
UpdateTime	DATETIME	NOT NULL	Timestamp when tracking updated.
Remarks	TEXT	NULL ALLOWED	KEEP EMPTY. Fill ONLY if delay/breakdown.
UpdatedBy (FK)	INT	NOT NULL	UserID of the staff who pushed update.

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7. LR_BILTY TABLE (Primary Contract)

The central pillar.

Column Name	Data Type	Constraint	Null Rules & Details
LrNumber (PK)	VARCHAR(30)	NOT NULL	Manual/Auto ID.
LrDate	DATE	NOT NULL	Date receipt generated.
ConsignorID(FK)	INT	NOT NULL	Sender ID from Customers.
ConsigneeID(FK)	INT	NOT NULL	Receiver ID from Customers.
Origin	VARCHAR(100)	NOT NULL	Start location.
Destination	VARCHAR(100)	NOT NULL	End location.
MaterialDesc	VARCHAR(200)	NOT NULL	Goods detail.
TotalWeight	DECIMAL(10,2)	NOT NULL	Package weight.
TotalPackages	INT	NOT NULL	Count of boxes.
FreightAmount	DECIMAL(10,2)	NOT NULL	Total fare calculated.
AdvanceAmt	DECIMAL(10,2)	NULL ALLOWED	EMPTY if no advance. Fill if paid.
BalanceAmt	DECIMAL(10,2)	NULL ALLOWED	EMPTY if no balance. Fill if pending.
VehicleID (FK)	INT	NULL ALLOWED	EMPTY if truck is missing. Fill when assigned.
DriverID (FK)	INT	NULL ALLOWED	EMPTY if driver is missing. Fill when assigned.
EWayBillNo	VARCHAR(20)	NULL ALLOWED	EMPTY until Govt Bill generated.
DeliveryDate	DATETIME	NULL ALLOWED	EMPTY while tracking. Fill when completed.
Status	VARCHAR(30)	NOT NULL	In Transit, Delivered.

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8. EWAY_BILL TABLE

Mandatory government data link.

Column Name	Data Type	Constraint	Null Rules & Details
EWayBillNo(PK)	VARCHAR(20)	NOT NULL	Official govt generated number.
EWayBillDate	DATETIME	NOT NULL	Date and time generated.
LrNumber (FK)	VARCHAR(30)	NOT NULL	Connects to LrBilty.
GenByGSTIN	VARCHAR(15)	NOT NULL	GST number of the generator.
SupplierGSTIN	VARCHAR(15)	NOT NULL	GST number of the sender.
RecipientGSTIN	VARCHAR(15)	NOT NULL	GST number of the receiver.
DispatchFrom	VARCHAR(200)	NOT NULL	Pincode it left from.
ShipTo	VARCHAR(200)	NOT NULL	Pincode it is going to.
Distance	INT	NOT NULL	Required transit distance (KM).
TotalInvoiceVal	DECIMAL(10,2)	NOT NULL	Total value of cargo.
CustomDutyVal	DECIMAL(10,2)	NULL ALLOWED	EMPTY if not cross-border shipping.
CGSTAmount	DECIMAL(10,2)	NULL ALLOWED	EMPTY/0 if tax exempt. Fill if standard rules.
SGSTAmount	DECIMAL(10,2)	NULL ALLOWED	EMPTY/0 if tax exempt. Fill if standard rules.
IGSTAmount	DECIMAL(10,2)	NULL ALLOWED	EMPTY/0 if tax exempt. Fill if standard rules.
VehicleNumber	VARCHAR(20)	NOT NULL	Reg number recorded per govt filing.
TransporterID	VARCHAR(50)	NULL ALLOWED	EMPTY if internal fleet. Fill if outsourced.
ValidUpto	DATETIME	NOT NULL	Government expiration point.